

Wenting Liu



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SUMMARY

PhD candidate in Computational Medicine and Bioinformatics at the University of Michigan, specializing in multimodal AI for clinical decision support. Experienced in developing scalable machine learning and LLM-based systems for real-world healthcare data, including clinical notes, physiological time-series, and medical imaging.

Core strengths include sequence modeling (RNNs, Transformers, Mamba), multimodal learning, and LLM-driven pipelines, with a focus on translating models into clinically meaningful and deployable solutions.

Proven track record in high-impact biomedical AI projects, including early risk prediction from ventilator waveforms (IEEE EMBC 2025, top 7%), LLM-based reconstruction of diagnostic reasoning, and drug combination modeling for cancer therapy.

Interested in advancing AI applications for real-world decision-making in healthcare and beyond, with emphasis on multimodal integration, model efficiency, and scalable deployment.

SKILLS

- **Programming & Data:** Python, Bash, SQL, R, MATLAB, C++
- **Machine Learning & Deep Learning Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers (LLMs, LoRA fine-tuning), PyTorch Geometric (PyG), scikit-learn; NumPy, SciPy, pandas
- **AI Models:** Sequence Modeling (RNN, LSTM, GRU), Transformers, State Space Models (Mamba), Large Language Models (LLMs), Multimodal Learning (Vision-Language Models, LoRA fine-tuning), Learning Paradigms (Learning Under Privileged Information (LUPI), Knowledge Distillation), Convolutional Neural Networks (CNNs: EfficientNet, U-Net), Reinforcement Learning (DQN), Graph Neural Networks (GCN, GraphSAGE).

EDUCATION

PhD Candidate in Computational Medicine and Bioinformatics

2024 – Present

University of Michigan, Ann Arbor, MI, USA 48109

Master of Science in Data Science

2021 – 2023

Michigan State University, East Lansing, MI, USA 48864

Bachelor of Engineering in Food Science and Technology

2007 – 2011

Ocean University of China, Qingdao, Shandong, China

WORK EXPERIENCE

BCIL Lab, Department of Computational Medicine and Bioinformatics, University of Michigan

Graduate Student Research Assistant | Ann Arbor, MI | Aug. 2024 – Present

- Lead research on applying AI and multimodal learning to real-world clinical data, focusing on clinical reasoning reconstruction and early risk prediction in critical care settings.
- Design and develop scalable machine learning and LLM-based systems for processing unstructured and time-series clinical data, including clinical notes and high-frequency physiological signals.
- Build end-to-end pipelines integrating large language models, sequence models, and multimodal learning approaches to support clinically meaningful decision-making.
- Collaborate with clinicians to translate domain knowledge into model design, enabling development of practical and scalable AI solutions for healthcare applications.

Department of Computational Medicine and Bioinformatics, University of Michigan

Graduate Student Instructor | Ann Arbor, MI | Aug. 2025 – Dec. 2025

- Led Python-based laboratory sections for BIOINF 580 (Signal Processing and Machine Learning in Biomedical Sciences), teaching core concepts in time–frequency analysis, digital filtering, feature extraction, and supervised learning for biomedical data.
- Developed ECG- and EEG-based instructional materials and reproducible coding pipelines to support hands-on learning in physiological signal analysis.
- Mentored interdisciplinary graduate students in applied machine learning, recognized for strong instructional clarity, organization, and effectiveness.

BCIL Lab, Department of Computational Medicine and Bioinformatics, University of Michigan

Research Associate | Ann Arbor, MI | Sep. 2023 – Aug. 2024

- Led development of machine learning models to predict cancer drug combination synergy using the

NCI-ALMANAC dataset, modeling dose-dependent effects across 100+ anticancer agents.

- Designed cell line-specific predictive frameworks capturing nonlinear interactions between drug combinations and dosage, improving identification of synergistic treatment strategies.
- Demonstrated the importance of incorporating dosage as a key factor in combination therapy modeling, with results published in IEEE CSCE 2024.

Garmire Lab, Department of Computational Medicine and Bioinformatics, University of Michigan

Research Assistant | Ann Arbor, MI | May. 2022 – Aug. 2023

- Co-first author on an epigenome-wide association study (EWAS) of severe preeclampsia, analyzing cord blood DNA methylation data to identify disease-associated epigenetic signatures.
- Developed a robust analytical workflow to improve estimation of cell type composition, enhancing the accuracy and interpretability of epigenetic associations in cord blood studies.
- Contributed to study design, data processing, and statistical analysis, with findings accepted for publication in GigaScience.

Corporate Planning Headquarters, Tsingtao Brewery Co., Ltd.

Senior Statistical Analyst & Process Optimization Lead | Qingdao, Shandong, China | Aug. 2011 – Nov. 2021

- Led data-driven process optimization initiatives across manufacturing operations, applying statistical analysis and Six Sigma methodologies to improve production efficiency and product quality.
- Delivered company-wide training in applied statistics, quality control, and process improvement, supporting implementation across ~70 manufacturing facilities.
- Drove adoption of data-informed decision-making and standardized analytical practices to enhance operational consistency and system-level performance.

PROJECTS

LLM-Based Reconstruction of Pulmonary Embolism (PE) Diagnostic Reasoning

- Developed LLM-driven pipelines to reconstruct clinical reasoning for pulmonary embolism (PE) from emergency department notes, combining structured feature extraction and summarization to convert unstructured text into clinically meaningful representations.
- Designed schema-guided prompting strategies and evaluated multiple LLMs (GPT-5 series, Mistral) for scalable, consistent abstraction of reasoning-relevant features from clinical narratives.
- Incorporated retrieval-augmented generation (RAG) using national and institution-specific clinical guidelines to improve reasoning grounding and reduce variability.

- Demonstrated feasibility of automated, auditable reconstruction of diagnostic processes across large patient cohorts, reducing reliance on manual chart review and enabling scalable clinical evaluation.

AI-Driven Risk Prediction for Pediatric Acute Respiratory Distress Syndrome (PARDS)

- Developed a multi-stage modeling framework for early risk prediction of PARDS using high-frequency mechanical ventilation signals from PICU patients.
- Established baseline predictive performance using traditional machine learning with advanced signal processing (DTCWPT), demonstrating feasibility of waveform-based early risk stratification. (IEEE EMBC 2025, top 7%)
- Built advanced sequence models (Transformer, Mamba, RNN-based architectures) to predict 12-hour oxygenation risk using first-hour ventilator data, capturing temporal dynamics of patient trajectories.
- Extended the framework toward multimodal learning by integrating ventilation signals and chest X-ray (CXR), leveraging Vision Transformers (ViT) with LoRA fine-tuning and Learning Under Privileged Information (TRAM) to enhance prediction performance.

Cancer Drug Combination Synergy Prediction (NCI-ALMANAC)

- Built predictive models for cancer drug combination synergy across multiple cell lines using large-scale pharmacogenomic data (NCI-ALMANAC).
- Applied neural network-based approaches to model dosage-sensitive interactions between drug pairs, capturing nonlinear synergy effects.
- Demonstrated improved prediction of synergistic growth inhibition and highlighted the importance of dose-aware modeling in combination therapy design.
- Results published in IEEE CSCE 2024, supporting the application of machine learning in drug discovery and combination therapy optimization.

Radiology Report Generation with Vision-Language Models (LLaMA Vision 11B)

- Fine-tuned a 4-bit quantized vision-language model (LLaMA 3.2 Vision 11B) to generate radiology-style reports from medical images (X-ray, CT, ultrasound).
- Implemented parameter-efficient fine-tuning (LoRA) across vision and language components, optimizing training efficiency under GPU memory constraints.
- Achieved significant improvements in generation quality (BLEU +20–44%, BERTScore +90%), demonstrating effective alignment between visual inputs and clinical text generation.
- Addressed key challenges in multimodal modeling, including image-text alignment, instruction sensitivity, and efficient large-model training.

Skin Lesion Classification with EfficientNet (MedMNIST)

- Developed deep learning models for multi-class skin lesion classification using dermoscopic images (MedMNIST dataset).
- Built end-to-end training pipelines with data preprocessing, augmentation, and optimization strategies (cosine learning rate scheduling, early stopping, checkpointing).
- Achieved strong classification performance ($AUC > 0.93$), with ensemble models improving robustness and generalization across classes.

Reinforcement Learning for HIV Treatment Optimization (HealthGym v2.0)

- Developed a Deep Q-Network (DQN) framework to optimize treatment strategies for HIV using longitudinal patient simulation data (~500K records).
- Modeled clinical decision-making as a sequential optimization problem, incorporating patient state dynamics (viral load, CD4) and treatment actions.
- Designed reward functions balancing viral suppression and immune recovery, enabling learned policies to outperform baseline strategies (avg return ~6.7 vs ~2.5).

Graph Neural Network Modeling and Optimization (PyTorch Geometric)

- Built and evaluated graph neural network models for node classification, learning representations from relational graph structures.
- Compared multiple architectures (GCN, GraphSAGE, APPNP) and improved performance through systematic hyperparameter tuning and model evaluation.
- Enhanced model reliability by identifying and resolving issues in the evaluation pipeline, improving overall model performance and reproducibility.

PUBLICATIONS

1. **Liu W**, et al. *Deep Learning on Mechanical Ventilation Signals for Early Prediction and Risk Stratification of PARDS*. Submitted to IEEE Transactions on Biomedical Engineering (TBME, special issue; under review).
2. **Liu W**, et al. *Mechanical Ventilation-Driven Machine Learning for Predicting Pediatric Respiratory Failure*. IEEE EMBC 2025 (top 7% of submissions).
3. **Liu W**, et al. *Predicting Efficacy of Cancer Drug Combinations Using Machine Learning*. IEEE CSCE 2024.
4. Yang X*, **Liu W***, et al. *Epigenome-wide analysis of severe preeclampsia*. Accepted in GigaScience.